1 What the strategy does

The Gold/Silver Reversion signal trades gold futures against a move that gold has made relative to silver. You run it on a gold chart with silver as the second data series. Silver is never traded. It is the reference that tells the signal whether gold's move was gold's own or part of a move in both metals.

1.1 The signal

Every bar the signal measures how far gold has moved over the last `WinBars` bars and subtracts `HedgeRatio` times silver's move over the same bars. The result is the relative move. It divides the relative move by its normal size, taken from the last `VolSamples` such moves, and calls that `z`. When `z` reaches `ZEntry` the signal sells gold. When `z` reaches minus `ZEntry` it buys gold. One setup per session, at market on the next bar.

After a trade the measuring window starts again on the exit bar and grows back to `WinBars`. No bar from inside a finished trade is measured twice.

1.2 The exits

There is no end-of-day flatten in this build. A position stays open until one of these closes it:

- **Stop** - gold has moved `GoldStopSd` session-sigmas against you since the signal bar.
- **Target** - gold has moved `GoldTpSd` session-sigmas in your favour since the signal bar.
- **Trail** - optional; off by default. See section 4.3.

One session-sigma is the usual size of a one-session gold move, taken from the last `RallySamples` sessions. A stop of `2.0` is therefore about two normal sessions of gold against you, whatever the year.

> **An open position blocks every new entry.** With no flatten, a trade that never reaches its stop or its target can sit for weeks and the signal takes nothing else meanwhile. Watch the open trade, not the entry count.

2 Setting it up

The signal needs two data series on one chart and a bar count large enough for its samplers. Everything else is in the inputs. The chart it expects:

Group	Gold relative-move fade, multi-day holds
Instrument	GC (QGC#C)
Data 1	5 Min (execution and signal)
Data 2	5 Min QSI#C (reference)
Time zone	CT. Defaults are set for an Exchange (Chicago) chart.

Step 1. Build the chart. Gold as Data1, silver as Data2, both 5-minute, both on the same session template, chart time zone set to Exchange (Chicago). The signal reads `Close data2` for silver; if silver is missing or on a different resolution the numbers are wrong without any error message.

Step 2. Set the bars back. In the strategy properties set **Maximum bars back** to at least `WinBars` plus `RallyBars` plus `ExtBars` plus five, or leave it on auto-detect. The default inputs need 365.

Step 3. Load the signal. Compile `GoldSilverReversion_NoEOD_FINAL`, add it to the chart, and check that the input list matches section 3.

Step 4. Run once before you change anything. The first `VolSamples` windows and the first `RallySamples` sessions are warm-up. No trade appears before about 50 sessions of data.

> **Changing the bar size changes every bar-count input.** `WinBars` 36 is three hours on 5-minute bars and 36 minutes on 1-minute bars. Convert `WinBars`, `RallyBars` and `ExtBars` together or the signal is a different signal.

3 The inputs

These are the defaults in the FINAL build. Inputs marked off are kept in the code for testing and do nothing at `-1` or `false`.

3.1 Signal

The inputs that decide when a setup fires:

INPUT	DEFAULT	WHAT IT DOES
`WinBars`	`36`	Length of the measuring window in bars.
`ZEntry`	`2.0`	How unusual the relative move must be before a trade is taken.
`HedgeRatio`	`5.0`	Weight of silver's move inside the relative move. See section 4.
`BetaDays`	`0`	`0` uses `HedgeRatio` as given. A number above 1 estimates it from that many sessions instead.
`VolSamples`	`300`	How many past windows define a normal relative move.
`ShortNeedsGoldUp`	`true`	A short is only taken when gold itself rose over the window.
`SkipBlocksDay`	`true`	A refused signal uses up the session's one entry.
`MaxEntriesDay`	`1`	Entries per session. Anything above 1 does nothing without an intraday exit.

`ZEntry` is measured in whatever unit `HedgeRatio` produces, so the two are read together.

3.2 Exits

The inputs that decide how a trade ends. All distances are in session-sigmas of gold:

INPUT	DEFAULT	WHAT IT DOES
`GoldStopSd`	`2.0`	Stop distance. `-1` is off.
`GoldTpSd`	`4.0`	Target distance. `-1` is off.
`GoldTrailMode`	`3`	1 fixed distance, 2 breakeven, 3 fraction of the best point.
`GoldArmSd`	`-1`	The trail switches on once the trade is this far in your favour. `-1` keeps the trail off.
`GoldTrailSd`	`1.0`	Mode 1 only: how much may be given back.
`GoldTrailFrac`	`0.5`	Mode 3 only: what share of the best point may be given back.
`RallyBars`	`288`	Bars in one session, the base of the session-sigma.
`RallySamples`	`100`	How many sessions define a normal session move.

The stop is the only exit that ends a losing trade. Leave it on.

3.3 Filters, window and size

Inputs you rarely touch:

INPUT	DEFAULT	WHAT IT DOES
`RallyFlipSd`	`-1`	Turns a long after a gold rally into a short. Measured worse than off.
`LongMaxRallySd`	`-1`	Drops a long after a gold rally. Measured worse than off.
`GoldExtSd`	`-1`	Requires gold itself to be extended against the trade. Measured worse than off.
`LongNeedsGoldDown`	`false`	Mirror of `ShortNeedsGoldUp` for longs. Measured worse than off.
`ExtBars`	`36`	Window for `GoldExtSd`. Unused while that filter is off.
`StartCT`, `NoEntryAfterCT`	`1700`, `1530`	Entry window in Chicago time. It wraps midnight.
`AnchorCT`	`1700`	Session start for the daily counter and the beta estimate.
`Dir`	`1`	`-1` reverses both sides. A control run, not a setting.
`PrintEvents`	`false`	Writes nothing in this build.
`NumContracts`	`1`	Contracts per entry.

Everything in this last table stays as delivered unless you are running a control.

4 What happens when you move an input

Every sweep below was run on 2017 to 2022 on 5-minute bars. The behaviour repeats across all the versions tested; the exact numbers do not, so none are given here.

4.1 HedgeRatio

`HedgeRatio` decides what the signal is actually measuring. Silver moves about 1.7 times as much as gold, so its weight matters more than the number suggests.

- `0.25` **to** `0.75` - the relative move is gold's move net of its usual co-movement with silver. This is the version where you fade gold's own excess move.
- `1.0` - silver already dominates. The relative move is mostly minus silver.
- **above** `2` - gold's own move is a small correction. `z` is silver's three-hour extreme with the sign turned round, and the trade is gold following silver back. This is a lead-lag trade, not a fade of the gold-silver relationship.
- `5` - gold contributes about a fifth of the signal. At `20` it contributes nothing; use `20` as the control run. If the trades barely change, gold plays no part in the entry.

In every sweep the average result rose towards the top of the `HedgeRatio` range. That is this effect, and at high values the long side tends to be long gold in metals rallies.

> **Do not read a high HedgeRatio as a better hedge.** Above `1` you are no longer hedging gold's move, you are trading silver's move with a gold contract.

4.2 ZEntry, WinBars and VolSamples

These three set how often the signal fires and how big the moves it fires on are.

- `ZEntry` **lower** - more trades, but the added ones sit just above the threshold, and between `1.5` and `2.0` they lost money in every version tested. Higher: fewer and larger divergences.
- `WinBars` **shorter** - faster bursts qualify and each is worth less; below about 12 bars the trades sit at the cost floor. Longer: rarer, larger, slower to react.
- `VolSamples` **fewer** - the yardstick follows the recent weeks, so a stress week lifts the bar for the next one. More: a stable yardstick. Between `100` and `500` little changes.

`ZEntry` is measured in the unit that `HedgeRatio` produces. `2.0` at `HedgeRatio` `5` is a silver-sigma threshold and is not comparable to `2.5` at `0.75`.

4.3 The exits

All three exits measure gold's move since the signal bar in session-sigmas. They do not look at silver and they do not look at the relative move.

- `GoldStopSd` **tight (**`1.0` **to** `1.5`**)** - cuts adverse moves that mostly came back; with an end-of-day flatten every level below 3 lowered the average trade. Without a flatten a tight stop is what keeps the trade count up, because it is the only thing that ends a losing trade.
- `GoldStopSd` **wide (**`3` **and above)** - touches single trades. Without a flatten one open loser can block entries for weeks.
- `GoldTpSd` - in the no-flatten sweeps `4` was the level that held across stop settings. `2` and `3` cut the multi-day winners short; `5` and `6` are rarely reached.
- **Trail, mode 3 with** `GoldArmSd` `1.0` **and** `GoldTrailFrac` `0.3` - roughly halved the drawdown for about a third of the profit. No trail setting beat no trail on profit.
- **Trail, mode 2** - only acts when armed late and then does nothing.

4.4 The two side rules

`ShortNeedsGoldUp` and `SkipBlocksDay` decide which shorts you take and what happens to a refused signal.

- `ShortNeedsGoldUp` **on** - at `HedgeRatio` `0.75` it removes about a third of the shorts, the ones where silver fell harder than gold. At `HedgeRatio` `1` and above it removes most shorts, because `z` reaches the short threshold mainly when both metals dropped.
- `SkipBlocksDay` **off** - a later crossing the same day trades instead. Those replacement trades were the losing population every time they were measured.

> **Check the long/short split after every change to HedgeRatio.** A run with almost no shorts is not a mean-reversion strategy any more; it is a long gold position triggered by silver.

4.5 The entry window

Entries after the Globex open, 17:00 to 02:00 Chicago time, had the highest average per trade. The US session added many small positive trades that fill the quiet stretches. Cutting the daytime raised the average and the drawdown together, so the full window stays.

4.6 Running without the end-of-day flatten

The FINAL build has no flatten. Compared with the intraday version this changes three things at once: the trade count falls sharply, because a position blocks entries for its whole life; the profit and the loss per trade grow by an order of magnitude; and the result becomes a multi-day exposure to gold's direction. Settings that looked stable on 2017 to 2022 in this mode did not repeat on 2023 to 2026.

5 What was tested and dropped

Each line is one idea, what it changed, and why it is not in the FINAL build.

- **Level spread against a rolling mean** - earned in quiet years but ran long-heavy at the hedge ratio it needed; same regime pattern as the burst signal.
- **Kalman-filter spread** - adapts the hedge ratio bar by bar; ranked third of the entry statistics tried.
- **Range position** - where the spread sits in its recent high-low range; noisy, same regime pattern.
- **MAD scale and percentile threshold** - neither beat the standard deviation.
- **Stall entry** - wait for a retrace before entering; traded count for average, profit unchanged.
- **Reversal-bar confirmation** - fitted the test years and failed the later ones.
- **Volume and volatility gates at entry** - cost more than half the profit.
- **Filters on gold's own position** - the longs entered into gold rallies were on average the better longs; every filter that removed them cost money.
- **Rally flip** - turning those longs into shorts was worse in every cell.
- **Spread-measured stops and targets** - the spread's excursions were unrelated to the profit of the trade; the largest adverse spread moves were the best trades.
- **Fixed-dollar stops and targets** - every level lowered the average; the profit was in the tail.
- **Time exit** - closing a trade that is not ahead after 15 bars removed winners as well as losers.
- **Silver-measured stops and targets** - best setting was off.
- **Evening-only window** - higher average, half the profit, no drawdown benefit.

6 Open items

- `HedgeRatio 20` **control** - if the result is unchanged, gold's own move plays no part in the signal.
- **Long/short split of the FINAL build** - on both periods, to see how much of the result is long gold.
- **Drawdown per contract** - one contract against a prop-firm limit is a sizing question that the signal does not answer.